SEAN MULHERIN

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PROFILE

Ph.D. candidate in Statistics at UCLA and Lead AI Architect with experience translating statistical research into robust production-grade systems. Research centers on spatial-temporal point processes, infectious disease dynamics, causal clustering, and statistics education. I bring seven years of secondary mathematics/science teaching, university-level instruction, and technical industry leadership; all of which working to solve complex quantitative problems.

RESEARCH INTERESTS

Spatial-temporal statistics; stochastic processes; infectious disease dynamics; statistics and data science education

EDUCATION

University of California, Los Angeles (UCLA) — Los Angeles, CA
Ph.D. in Statistics, expected 2028 — *03/2025 – Present*

– Advisor: Prof. Frederic Schoenberg.

University of California, Los Angeles (UCLA) — Los Angeles, CA
M.S. in Applied Statistics and Data Science — *09/2023 – 03/2025*

– Thesis: *Testing for Causal Clustering in K-12 Student Discipline*.
– Advisor: Prof. Frederic Schoenberg.
– Honors: UCLA Statistics Outstanding Masters Student Award, 2025.

University of North Carolina at Chapel Hill — Chapel Hill, NC
M.A.T. in Secondary Mathematics — *05/2019 – 09/2020*

– Advisor: Dr. Josh Corbat.

North Carolina State University — Raleigh, NC
B.S. in Mathematics — *09/2015 – 05/2019*

CORE COMPETENCIES & TECHNICAL SKILLS

– **Statistics & Machine Learning:** Spatial-temporal modeling, point processes, multivariate statistics, experimental design, causal inference, forecasting, neural networks, elastic net regularization, model validation, and NLP pipelines.
– **Programming & Engineering:** Python, R, SQL, JavaScript, HTML, Git/GitHub, parallel computing, web-based model integration, and reproducible research workflows.
– **Production & Cloud:** Scalable AI architecture, semantic similarity analysis, confidence scoring, clustering, deduplication, AWS, GCP, Apache Airflow, and Docker.

PROFESSIONAL & ACADEMIC EXPERIENCE

Blueblade Los Angeles, CA
Lead AI Architect *06/2025 – Present*

- Architected and deployed a production-grade, multi-stage AI pipeline that transforms unstructured government regulations into validated, machine-readable compliance outputs, processing more than 5,000 documents monthly and accelerating manual processing by over 600%.
- Engineered NLP systems using statistical reasoning and semantic similarity analysis, improving extraction accuracy to 96% for more than 10 enterprise users.
- Designed confidence scoring, clustering, deduplication, and validation frameworks to manage uncertainty in a high-stakes regulatory setting, reducing compliance errors by 33%.
- Lead a cross-functional team of engineers and data scientists and oversee integration of AI models into scalable applications with Python back ends and JavaScript front ends.

UCLA Department of Statistics and Data Science Los Angeles, CA
Graduate Student Researcher, Teaching Fellow, and Student Mentor *09/2023 – Present*

- Conduct a text-as-data study of 27,879 open-ended teacher professional-development responses, applying transformer embeddings, clustering, and semantic dispersion analysis to examine changes across PD days and cohorts.
- Teach and mentor more than 300 students across Introduction to Statistical Reasoning, Societal Impacts of Data, Data Technologies for Data Scientists, multivariate statistics, and parallel computing in R.
- Serve as a student ambassador and UCLA Statistics Club mentor, translating advanced statistical concepts for technical and non-technical audiences.

Secondary Education Institutions (Brentwood School, Jackson Hole, etc.) Various Locations
Mathematics & Physics Department Faculty / Academic Advisor *08/2019 – 08/2025*

- **Technical Leadership & Instruction:** Directed quantitative curriculum development across multiple institutions, training students in advanced mathematics (Calculus, Statistics, Trigonometry, Geometry, Algebra) and Physics.

PUBLICATIONS & SCHOLARLY WORK

- Mulherin, S., and Schoenberg, F. (2026). "Mitigating Spurious Cross-Excitation in Spatial-Temporal Hawkes Models of Infectious Disease Transmission." Manuscript submitted to *The Annals of Applied Statistics*, July 2026.
- Mulherin, S. (2025). *Testing for Causal Clustering in K-12 Student Discipline*. M.S. thesis, University of California, Los Angeles.

RESEARCH EXPERIENCE

Ph.D. Dissertation Research In progress

Covariate-Informed Estimation of Cross-Productivity in Spatiotemporal Hawkes Models of Infectious Disease

- Advisor: Prof. Rick Schoenberg.
- Develop a staged estimation framework that separates local and cross-location excitation in multivariate Hawkes models of infectious-disease incidence.
- Introduce row-mass conservation, distance- and mobility-weighted elastic-net penalties, and local-share regularization to mitigate spurious cross-productivity while preserving estimated local dynamics.
- Evaluate the proposed estimators through an ablation study spanning multiple cross-excitation regimes and an application to 2020 U.S. COVID-19 incidence.
- Improve held-out predictive performance relative to the joint baseline, reducing pooled RMSE by approximately 33%, while producing more interpretable productivity matrices.

UCLA Graduate Student Research 2026 – Present

Semantic Analysis of Teacher Professional-Development Feedback

- Advisor: Prof. Rob Gould
- Built a reproducible R pipeline to harmonize and analyze 27,879 open-ended Got and Need responses from teacher professional-development cohorts spanning 2016–2025.
- Compare RoBERTa, BERT, and sentence-transformer embeddings using PCA, k-means clustering, semantic centrality, and centroid-based dispersion to examine changes across PD days, question subjects, and cohorts.

UCLA Master's Thesis 2025

Testing for Causal Clustering in K-12 Student Discipline

- Committee: Prof. Rick Schoenberg (chair), Prof. Rob Gould, Dr. Nicolas Christou, Dr. Mike Tsiang,
- Applied likelihood-ratio tests based on information gain to compare Neyman-Scott and Hawkes models using disciplinary records grouped by day, school, and school year from 2016–2023.
- Found statistically significant evidence of causal triggering in seven of twelve school-year combinations, with conclusions strongly governed by event frequency and magnitude.

Advanced Studies Institute in Mathematics of Data Science & Machine Learning 2024

National Science Foundation-sponsored research program, Uzbekistan

- Awarded travel support for a specialized workshop covering model-based clustering, Hawkes processes, benign overfitting, generalization, double descent, and mirror descent.

University of North Carolina at Chapel Hill Research Project 2020

Assessing Collaboration and Critical Thinking Opportunities in Online Learning at the Secondary Level

- Collected survey and interview data from 36 secondary instructors during the COVID-19 transition to virtual learning and studied relationships among group work, participation, collaboration, and critical thinking.

North Carolina State University, College of Design 2017 – 2018

Research Assistant

- Collected data on relationships between healthy diets and elementary students' social, emotional, and academic performance.

TALKS & PRESENTATIONS

DataX Data Science Education Symposium	2026
UCLA STATS 147 Guest Speaker – Parallel Computing in R	2026
UCLA STATS 411 Guest Speaker – Multivariate Statistical Analysis	2026
International Conference on Infectious Disease Dynamics – Poster Presentation	2025
DataFest Conference Guest Speaker – Introduction to R	2024
DataFest Conference Guest Speaker – Data Cleaning and Wrangling in R	2024

AWARDS, SERVICE & ACTIVITIES

UCLA Statistics Outstanding Masters Student Award	2025
UCLA Math and Physical Sciences Council Member	2023 – 2025
National Institute of Statistical Sciences GSN Council Member	2023 – 2025
Collegiate Cross Country & Track Athlete	2015 – 2019